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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	XMM
Selected Domain	Domain 2 — Orbyte Smart SME Challenge
Selected SDG Track (1–17)	SDG 8 — Decent Work and Economic Growth (8.2, 8.3); relevant alignment with SDG 9.3
Project / Solution Name	MouManTAI
Name (Leader)	Wong Jia Hui
Email & Phone (Leader)	wogjiahui.0111@gmail.com / +6017-4662286
University Affiliation(s)	Universiti Teknologi Malaysia (UTM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia has 1.29 million MSMEs, which is 96.2% of all businesses, contributing 39.7% of GDP (SME Corp Malaysia; DOSM, 2025). Many of them still rely on WhatsApp, spreadsheets and manual follow-up to run daily operations.

Take a typical food or catering SME. One customer promise is scattered across three places:

- Chats: the order arrives on WhatsApp, often with missing details.
- Spreadsheets: stock, ingredient expiry and production capacity.
- Supplier chats: quotations and delivery delays.

Before saying “yes”, the owner has to rebuild the order by hand, chase missing details, and check whether the order can really be delivered. When a supplier is late, the owner has to work out which orders are affected, what to do about it, and whether the customer agreed to the change.

The result is slow replies, accepted orders that cannot be delivered, wasted perishable stock, and unclear responsibility when plans change. Enterprise ERP systems are too expensive and complicated for a five-person business.

MouManTAI solves one recurring problem: making and protecting reliable customer commitments. It helps SME owners, managers and staff, while customers and suppliers keep using the channels they already use.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

MouManTAI (Cantonese 冇問題, "no problem") is an AI operations layer that links Customer, SME and Supplier in one evidence-backed workflow: Understand → Decide → Anticipate → Recover → Learn.

Where AI is used. Simple rules handle known cases first. When a message is unclear, Qwen3 turns it into a structured order draft with source evidence, and GLiNER helps extract details. If Qwen fails or times out, the Gemini API acts as a backup. Every AI output is checked against the original message, and anything uncertain goes to Human Review. AI never calculates stock or approves actions.

Customer. Messages and attachments become order drafts. The system asks one targeted question for each missing detail, and sends the case to Human Review after two answered attempts without progress. Complete orders go to Pending Acceptance and are never auto-confirmed.

SME. A Command Center shows orders against stock, expiry, recipes (BOM) and capacity. One feasibility engine explains whether each order can be fulfilled. When orders compete for the same resources, Google OR-Tools CP-SAT compares First-Come-First-Served, Profit and Expiry-Aware plans, and the same engine re-checks every plan.

Supplier. Offers are checked against category specifications before price.

Emergency. When a supplier delay is confirmed, the system finds the affected orders, compares recovery options by cost and delay, and acts only after explicit SME approval.

Workshop skills applied: GenAI APIs (Qwen3 on RunPod, Gemini API fallback), database architecture (PostgreSQL with versioned, audited records), and developer tools (Docker Compose, GitHub Actions, Replit).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 19, TypeScript and Vite. One SME Command Center with pages for Inbox, Orders, Human Review, Calendar, Suppliers, Forecast, Recovery and Audit.

Backend: Python, FastAPI and Pydantic. One app split into clear modules, such as message reading, orders, feasibility, planning, suppliers, forecasting, recovery and communication.

Database: PostgreSQL with SQLAlchemy and Alembic. It stores commitments, approvals, original messages and the audit trail.

AI and planning: Qwen3-4B and GLiNER on a RunPod Serverless GPU endpoint. PaddleOCR and PyMuPDF read PDFs and images. Google OR-Tools CP-SAT plans competing orders. Simple moving-average and seasonal forecasts predict demand.

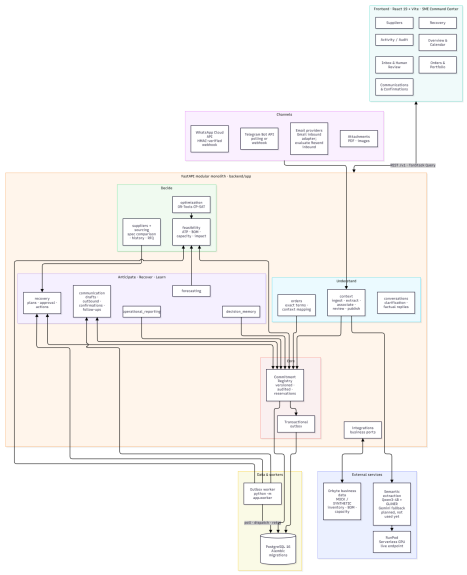
Channels and data: Telegram, WhatsApp Cloud API and Gmail bring in messages. Orbyte stock, recipe and capacity data come through a mock adapter.

Hosting: The React frontend runs on Vercel and the FastAPI backend runs on Render, both on free Hobby plans. Docker Compose is used for running everything locally. The database runs on Render Postgres.

System block diagram: A customer or supplier message arrives by Telegram, WhatsApp or Email. Simple rules read it first. If they cannot, GLiNER and Qwen3 on RunPod help. Clear orders go to Pending Acceptance, and unclear ones go to a clarification question or Human Review. The feasibility engine and OR-Tools check whether the order can be done. The SME accepts or rejects, and the decision is saved in PostgreSQL. Supplier delays, forecasting and recovery plans use the same records. Customers get replies from approved templates.

Sponsor resources: Yes. Our backend calls a live RunPod Serverless endpoint that runs Qwen3 and GLiNER to read messy messages. The SDC workshops taught us how to build and use it. Replit was our development workspace.

Block Diagram:
https://drive.google.com/file/d/1wZLCH7w1sx4F3cqQ6S19wTS_NsslhE0i/view?usp=sharing


Frontend · React 19 + Vite · SME Command Center
Suppliers
Recovery
Activity / Audit
Overview & Calendar
Inbox & Human Review
Orders & Portfolio
Communications & Confirmations
Channels
WhatsApp Cloud API HMAC-verified webhook
Telegram Bot API polling or webhook
Email providers SMART inbound adapters Postmark-compatible inbound
Attachments PDF · Images
REST /v1 + Realtime Query
FastAPI modular monolith · backend/app
Recipe
application DB Tools DB Ref
suppliers + inventory Tools DB Ref Tools DB Ref · ATS
Feasibility ATP · BOM · capacity · impact
Anticipate · Recover · Learn
forecasting
recovery plans · approval · actions
communication drafts · outbound confirmations · follow-ups
operational_reporting
decision_memory
Understand
orders · quotes · items · contact mapping
product ingest · extract · match · review · publish
conversations clarification · textual replies
Core
Canonical ledger · versioned · auditable · transactional
Transactional outbox
Integrations
Business ports
Data & workers
Outbox worker · retry · dead-letter
pgvector · migrations
External services
Alibaba Business Opportunity · RFQ2 / 1919TAT3C · Product API · inventory
Semantic Scholar · Search API · GLAMR · Unpaywall · planned, not used yet
Qwen/Kimi · Serverless LMs · live endpoint

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: We mapped how a typical small caterer handles a 100-lunch-box order across WhatsApp, spreadsheets and supplier chats, and listed the questions staff face: What exactly did the customer ask? Can we deliver it? Which supplier really fits? What changed, and who approves the fix?

Define: One problem: making and protecting reliable customer commitments. Not another chatbot or ERP.

Ideate & prototype: One SME Command Center answers those questions directly. Pending Acceptance is separate from Human Review. The calendar separates accepted from pending work. Optimisation shows trade-offs instead of one answer. Supplier comparison checks specifications before price.

Test: Automated tests in GitHub Actions check key rules: unclear orders go to a person, orders cannot be accepted after their terms change, plans are re-checked, and nothing runs without manager approval.

Deploy: The app runs on Docker Compose with PostgreSQL. We are using Vercel (frontend) and Render (backend). The server enforces tenant isolation and manager approval. If RunPod is down, rules still read the message and unclear ones go to Human Review. If the solver times out, single-order checks still work.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 8.2**: Achieve higher levels of economic productivity through technological upgrading and innovation. **SDG 8.3**: Support productive activities, entrepreneurship and the growth of micro-, small- and medium-sized enterprises. Supporting: **SDG 9.3**: Increase access of small-scale enterprises to financial services and integration into value chains and markets.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	• Small-business owners spend about 36% of a 45.5-hour week (≈16 hours) on administrative tasks [1]. • About 77% of Malaysian SMEs remain at the basic digitalisation stage [2]. • In a typical catering workflow, one order decision needs staff to check 3 disconnected sources by hand (customer chat, stock/capacity sheet, supplier chat), with no record linking the decision to its evidence. [1] Time etc, "The Big Price of Small Tasks" (2023). [2] SME Corp Malaysia; DOSM MSME Performance 2025.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Primary:** Order Decision Time, the median minutes from an order entering Pending Acceptance to the SME's accept/reject decision. **Secondary:** (1) Manual handling steps per order. (2) Risk detection time, from receiving verified supplier evidence

Metric Category	Your Response
	to the alert. (3) % of emergency actions executed with a recorded human approval.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Measured on team-created synthetic scenarios, not real SME data: • ≥30% shorter median Order Decision Time than the manual workflow on the same scripted tasks. • 100% of affected orders identified in scripted supplier-delay tests. • 100% of emergency actions human-approved before execution.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Qwen3-4B-Instruct-2507 (Alibaba Qwen)	Model	Apache-2.0
GLiNER multi v2.1 (urchade/gliner_multi-v2.1)	Model	Apache-2.0
PaddleOCR	Code and model	Apache-2.0
GLiNER library, Hugging Face Transformers	Code	Apache-2.0
PyTorch	Code	BSD-3-Clause
Google OR-Tools (CP-SAT)	Code	Apache-2.0
PyMuPDF	Code	AGPL-3.0
Python	Language	PSF License

FastAPI, Pydantic, pydantic-settings, SQLAlchemy, Alembic	Code	MIT
Uvicorn, httpx	Code	BSD-3-Clause
psycopg 3	Code	LGPL-3.0
RunPod Python SDK, pytest	Code	MIT
PostgreSQL 16	Database	PostgreSQL License
React, Vite, React Router, TanStack Query	Code	MIT
TypeScript	Code	Apache-2.0
openapi-typescript, Vitest, Testing Library	Code	MIT
Docker Compose	Tool	Apache-2.0
Datasets	None from outside. All data is synthetic data made by the team.	N/A

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

WONG

(SIGNATURE — WONG JIA HUI, TEAM LEADER)

25 October 2026